1.1. Môi trường prod:

Ta sẽ triển khai 3 máy ảo để tạo cụm gồm 1 máy manager và 2 máy worker và đảm bảo 3 máy ảo đã được cài đặt Docker Engine

1.1. Chỉnh hostname cho 3 máy ảo:

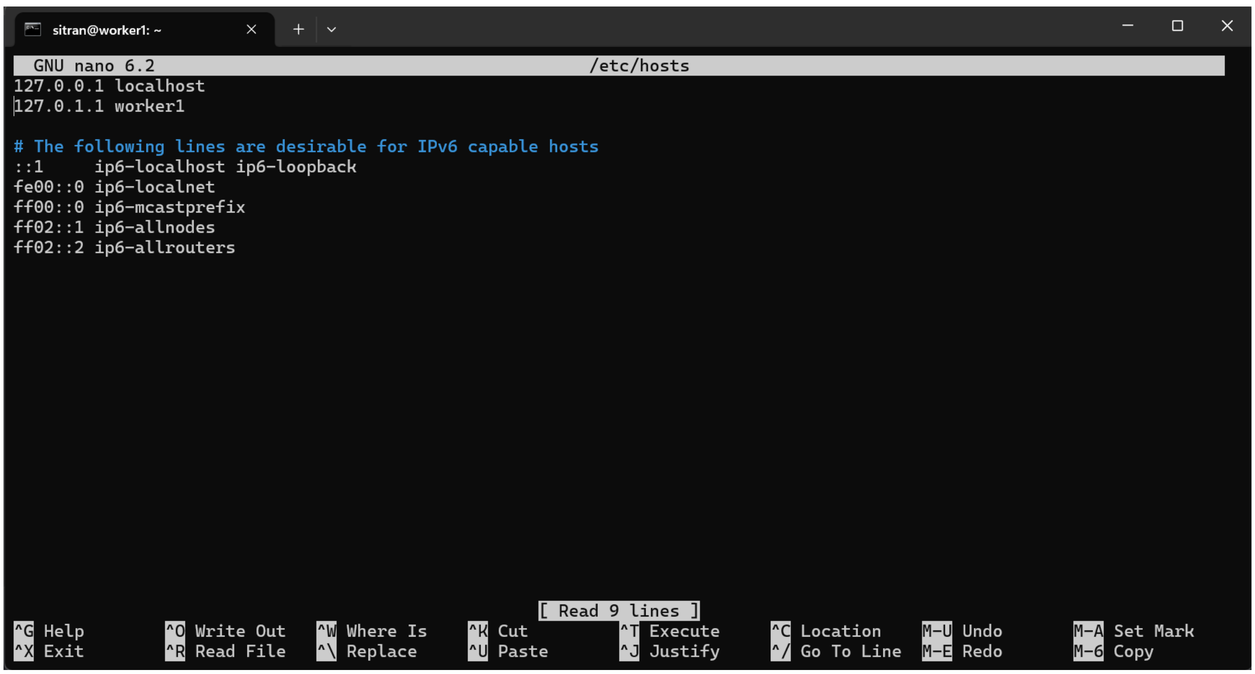
Giả định Hostname của các máy như sau:

- **Manager**: manager
- **Worker 1**: worker1
- **Worker 2**: worker2

Chạy lệnh:

```
sudo hostnamectl set-hostname <hostname>
```

Sau đó vào đường dẫn `sudo nano /etc/hosts` chỉnh tên kế bên IP 127.0.1.1



Sau đó chạy lệnh `exec bash` để áp dụng

1.2. Chỉnh IP tĩnh cho 3 máy ảo:

Giả định IP của các máy như sau:

- **Manager**: 192.168.1.10

- **Worker 1**: 192.168.1.11
- **Worker 2**: 192.168.1.12

Xác định tên card mạng với lệnh: `ip a`

```
sitran@worker1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:e0:ff:9c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.24/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
       valid_lft 86373sec preferred_lft 86373sec
    inet6 2001:ee0:51d6:dd00:20c:29ff:fee0:ff9c/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 2591974sec preferred_lft 604774sec
    inet6 fe80::20c:29ff:fee0:ff9c/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 06:9a:96:98:23:88 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

➔ Ens33

Các file cấu hình của Netplan nằm trong thư mục /etc/netplan/. Hãy kiểm tra tên file bằng lệnh:

`ls /etc/netplan/`

```
sitran@worker1:~$ ls /etc/netplan
50-cloud-init.yaml
```

Tắt cloud-init quản lý mạng (nếu không tắt, khi reboot máy có thể bị ghi đè lại file cũ)

`sudo bash -c 'echo "network: {config: disabled}" > /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg'`

```
sitran@worker1:~$ sudo bash -c 'echo "network: {config: disabled}" > /etc/cloud/cloud.cfg.d/99-disable-network-config.cf
g'
[sudo] password for sitran:
```

Mở file đó bằng trình soạn thảo nano:

`sudo nano /etc/netplan/50-installer-config.yaml`

```
 sitran@worker1: ~
  GNU nano 6.2                        /etc/netplan/50-installer-config.yaml *
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.1.11/24    # Thay IP này thành IP tĩnh muốn đặt
      routes:
        - to: default
          via: 192.168.1.1    # IP Gateway (Router/VMware Gateway)
      nameservers:
        addresses:
          - 8.8.8.8           # DNS Google
          - 1.1.1.1           # DNS Cloudflare

^G Help      ^O Write Out  ^W Where Is  ^K Cut      ^T Execute  ^C Location  M-U Undo  M-A Set Mark
^X Exit      ^R Read File  ^\ Replace   ^U Paste    ^J Justify  ^/ Go To Line M-E Redo M-6 Copy
```

Áp dụng cấu hình

```
sudo netplan try
```

```
sitran@worker1:~$ sudo netplan try
[sudo] password for sitran:

** (process:2087): WARNING **: 13:45:03.917: Permissions for /etc/netplan/50-installer-config.yaml are too open. Netplan
 configuration should NOT be accessible by others.

** (generate:2089): WARNING **: 13:45:03.921: Permissions for /etc/netplan/50-installer-config.yaml are too open. Netpla
n configuration should NOT be accessible by others.
WARNING:root:Cannot call Open vSwitch: ovsdb-server.service is not running.

** (process:2087): WARNING **: 13:45:04.143: Permissions for /etc/netplan/50-installer-config.yaml are too open. Netplan
 configuration should NOT be accessible by others.

** (process:2087): WARNING **: 13:45:04.296: Permissions for /etc/netplan/50-installer-config.yaml are too open. Netplan
 configuration should NOT be accessible by others.

** (process:2087): WARNING **: 13:45:04.296: Permissions for /etc/netplan/50-installer-config.yaml are too open. Netplan
 configuration should NOT be accessible by others.
Do you want to keep these settings?


Press ENTER before the timeout to accept the new configuration


Changes will revert in 118 seconds
Configuration accepted.
```

Kiểm tra xem IP đã đổi thành công chưa

```
ip a show ens33
```

```
sitran@worker1:~$ ip a show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:e0:ff:9c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.11/24 brd 192.168.1.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 2001:ee0:51d6:dd00:20c:29ff:fee0:ff9c/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 2591941sec preferred_lft 604741sec
    inet6 fe80::20c:29ff:fee0:ff9c/64 scope link
       valid_lft forever preferred_lft forever
```

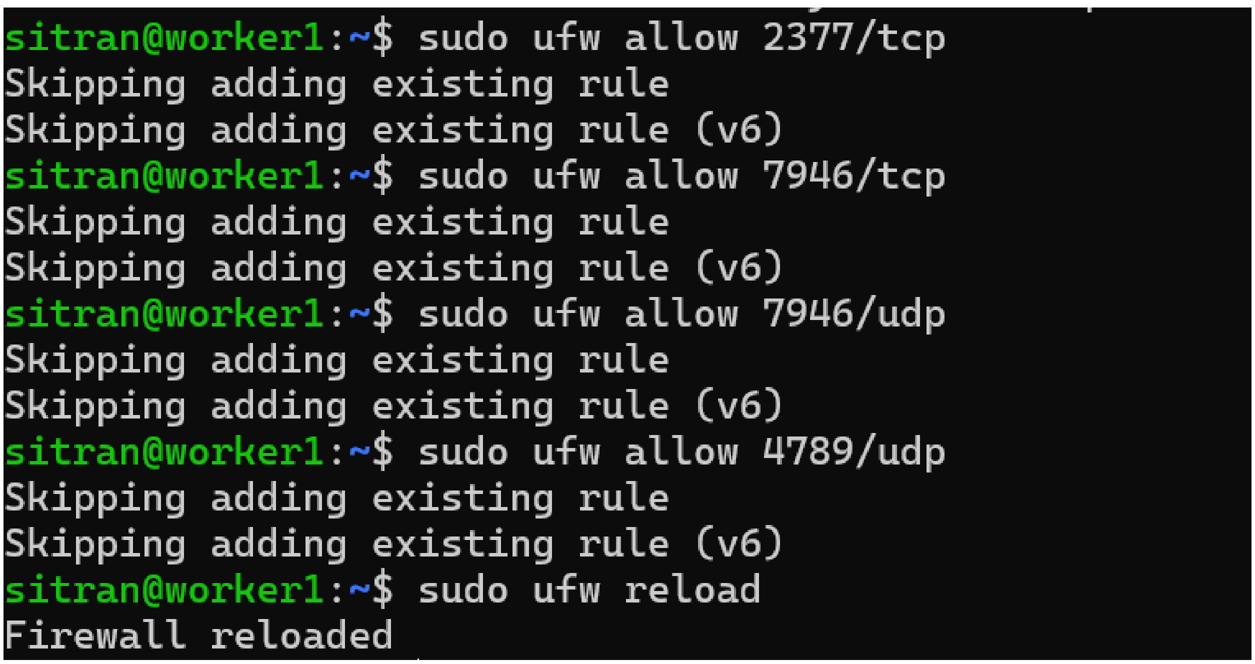
1.3. Mở cổng Firewall trên 3 máy ảo:

Docker Swarm cần một số cổng để các máy giao tiếp với nhau.

Nếu **không dùng UFW** (hoặc tắt ufw): Bỏ qua bước này.

Nếu **đang bật UFW**, chạy lệnh này trên **CẢ 3 MÁY**:

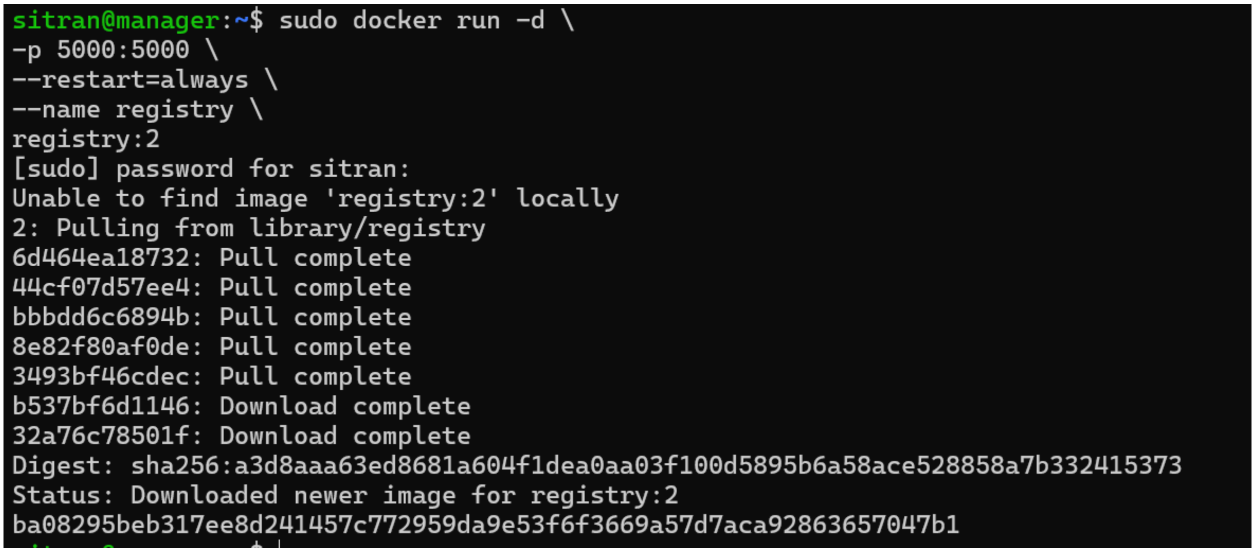
```
sudo ufw allow 2377/tcp

sudo ufw allow 7946/tcp

sudo ufw allow 7946/udp

sudo ufw allow 4789/udp

sudo ufw reload
```



1.4. Dựng registry nội bộ trên máy ảo manager:

- Thực hiện chạy dịch vụ lưu trữ image cục bộ trên cổng 5000:

```
sudo docker run -d \ -p 5000:5000 \ --restart=always \ --name registry
\ registry:2
```


```
sitran@manager:~$ sudo docker run -d \
-p 5000:5000 \
--restart=always \
--name registry \
registry:2
[sudo] password for sitran:
Unable to find image 'registry:2' locally
2: Pulling from library/registry
6d464ea18732: Pull complete
44cf07d57ee4: Pull complete
bbbdd6c6894b: Pull complete
8e82f80af0de: Pull complete
3493bf46cdec: Pull complete
b537bf6d1146: Download complete
32a76c78501f: Download complete
Digest: sha256:a3d8aaa63ed8681a604f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
ba08295beb317ee8d241457c772959da9e53f6f3669a57d7aca92863657047b1
```

- Cấu hình chấp nhận http:

Cấu hình ngoại lệ trên **cả 2 máy ảo** bằng cách tạo/chỉnh sửa file /etc/docker/daemon.json:

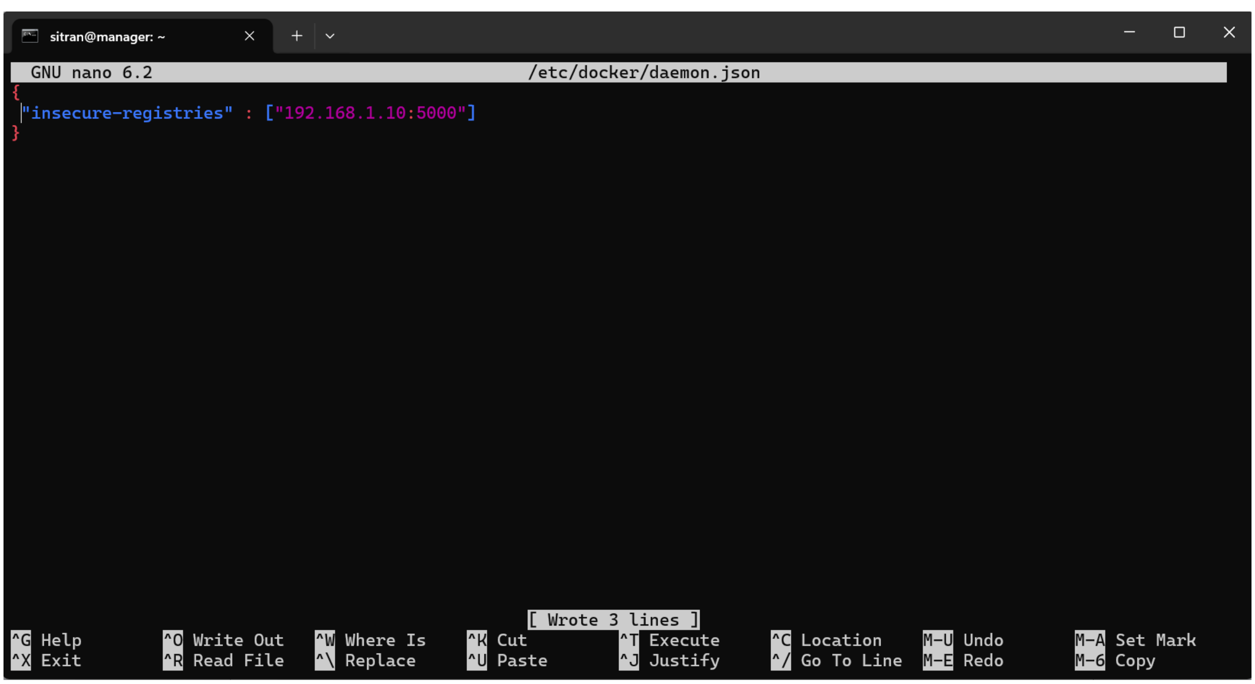
```
sudo nano /etc/docker/daemon.json
```

Thêm nội dung:

{

"insecure-registries" : ["192.168.1.15:5000"]

}


```
sitran@manager: ~
GNU nano 6.2                    /etc/docker/daemon.json
{
"insecure-registries" : ["192.168.1.10:5000"]
}

                    [ Wrote 3 lines ]
^G Help      ^O Write Out  ^W Where Is  ^K Cut    ^T Execute  ^C Location    M-U Undo  M-A Set Mark
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy
```

- Khởi động lại dịch vụ docker

```
sudo systemctl restart docker
```

- Đẩy image đã build trên máy cá nhân lên regitry vừa tạo:

Image phải có tag là <ip-server:port>/<image-name>:<version>

Dùng lệnh `docker push <image>`

```
PS C:\Users\trand> docker push 192.168.1.10:5000/pharmacy_backend:3.0
The push refers to repository [192.168.1.10:5000/pharmacy_backend]
b327eaca63b3: Waiting
b2e2e65549fe: Waiting
d73d92311bea: Waiting
f88d705613ca: Waiting
e640fc6efa14: Waiting
a9be9fd915e9: Waiting
2c1ce1d0a589: Waiting
4f4fb700ef54: Waiting
574a6ffeb08e: Waiting
2fd88c799806: Waiting
2c1ce1d0a589: Pushed
574a6ffeb08e: Pushed
d73d92311bea: Pushed
b327eaca63b3: Pushed
a9be9fd915e9: Pushed
b2e2e65549fe: Pushed
```

- Kiểm tra xem đã đẩy thành công chưa:

Curl http://192.168.1.10:5000/v2/_catalog

```
sitran@manager:~$ curl http://192.168.1.10:5000/v2/_catalog
{"repositories":["pharmacy_backend","pharmacy_frontend"]}
```

1.5. Khởi tạo cụm tại máy manager và thêm các worker:

- Khởi tạo cụm trên máy manager:

```
sitran@manager:~$ sudo docker swarm init \
--advertise-addr 192.168.1.10
[sudo] password for sitran:
Swarm initialized: current node (pdbqailj1z7o4srcxed03e2r9) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-021acn50rndaq9nmggjhlm64qpp9fee5510ki1oel8pdw8xour-cf33doxa4ed4q4ctcg54l9nc5 192.
168.1.10:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
sitran@manager:~$
```

- Join vào cụm ở 2 máy worker:

```
sitran@worker1:~$ docker swarm join --token SWMTKN-1-021acn50rndaq9nmggjhlm64qpp9fee5510ki1oel8pdw8xour-cf33doxa4ed4q4ct
cg54l9nc5 192.168.1.10:2377
This node joined a swarm as a worker.
```

```
sitran@worker2:~$ docker swarm join --token SWMTKN-1-021acn50rndaq9nmggjhlm64qpp9fee5510ki1oel8pdw8xour-cf33doxa4ed4q4ct
cg54l9nc5 192.168.1.10:2377
This node joined a swarm as a worker.
```

- Kiểm tra trên máy manager:

```
sitran@manager:~$ sudo docker node ls
[sudo] password for sitran:
ID                            HOSTNAME   STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
pdbqailj1z7o4srcxed03e2r9 *   manager    Ready     Active         Leader           29.6.0
pww6rol4xfina645hskxhv4t2     worker1    Ready     Active                          29.6.0
izrl4kzces6ehj0z3ukk5gtfq     worker2    Ready     Active                          29.6.0
```

- Tạo thư mục pharmacy và dùng win scp để kéo thư mục triển khai của prod và file compose base qua máy manager:

```
sitran@manager:/opt/pharmacy/prod$ ls
data  docker-compose.prod.yml  docker-compose.yml  nginx  secrets
```

File có dấu . phía trước đc linux mặc định xem là file ẩn

```
sitran@manager:/opt/pharmacy$ ls -a prod/
.  ..  data  docker-compose.prod.yml  .env.prod  nginx  secrets
```

1.6. Chạy cụm:

- Điền giá trị cho file .env.prod

```
sitran@manager: /opt/pharm

  GNU nano 6.2                          prod/.env.prod *
# Cấu hình container frontend
FRONTEND_IMAGE_TAG=2.0
DOCKER_REGISTRY_IP=192.168.1.10:5000

# Cấu hình container backend
SPRING_BOOT_IMAGE_TAG=3.0
MONGODB_DATABASE=pharmacy
SPRING_PROFILES_ACTIVE=prod

# Cấu hình container mongodb
MONGO_INITDB_ROOT_USERNAME=root
MONGO_INITDB_DATABASE=pharmacy

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

- Do chỉnh trên máy window cần:

```
sed -i 's/\r$//' .env.prod
```

```
sed -i 's/\r$//' docker-compose.yml

sed -i 's/\r$//' docker-compose.prod.yaml
```

- Chạy lệnh triển khai cụm:

Để chạy toàn bộ hệ thống, bạn sử dụng lệnh `docker stack deploy`.

Vì Swarm không tự đọc `.env`, chúng ta sẽ dùng một mẹo nhỏ là xuất (export) các biến từ tệp `.env` ra môi trường tạm thời ngay lúc chạy lệnh:

```
set -a && source .env.prod && set +a && docker stack deploy -c docker-compose.yml -c docker-compose.prod.yml pharmacy_app
```

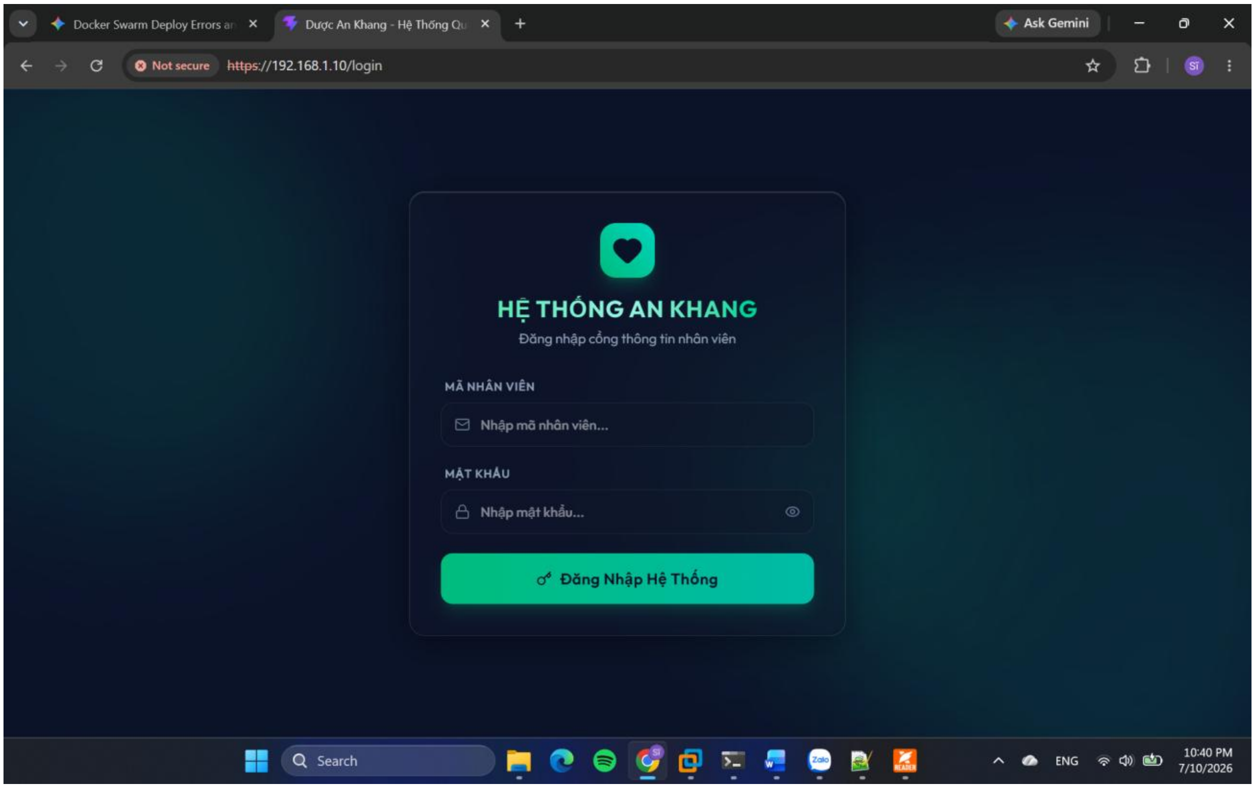
```
sitran@manager:/opt/pharmacy/prod$ set -a && source .env.prod && set +a && docker stack deploy -c docker-compose.yml -c docker-compos
e.prod.yml pharmacy_app
Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating secret pharmacy_app_mongo_uri
Creating secret pharmacy_app_domain_crt
Creating secret pharmacy_app_domain_key
Creating secret pharmacy_app_db_password
Creating secret pharmacy_app_jwt
Creating config pharmacy_app_nginx_config
Creating service pharmacy_app_spring-boot
Creating service pharmacy_app_frontend
Creating service pharmacy_app_mongodb
```

- Kiểm tra:

```
sitran@manager:/opt/pharmacy/prod$ sudo docker stack services pharmacy_app
ID             NAME                       MODE         REPLICAS   IMAGE                                       PORTS
som4rmzx1z42   pharmacy_app_frontend      replicated   3/3        192.168.1.10:5000/pharmacy_frontend:2.0   *:80->80/tcp, *:443->443/
tcp
5or41etfvgj1   pharmacy_app_mongodb       replicated   1/1        mongo:8
pp860l752sg2   pharmacy_app_spring-boot   replicated   3/3        192.168.1.10:5000/pharmacy_backend:5.0
```

```
sitran@manager:/opt/pharmacy/prod$ sudo docker stack ps pharmacy_app
ID             NAME                         IMAGE                                       NODE      DESIRED STATE   CURRENT STATE
        ERROR     PORTS
ken8a3ucb2ji   pharmacy_app_frontend.1      192.168.1.10:5000/pharmacy_frontend:2.0   worker2   Running         Running about a minut
e ago
ogmu8nckp5vg   pharmacy_app_frontend.2      192.168.1.10:5000/pharmacy_frontend:2.0   manager   Running         Running about a minut
e ago
tdmb1qabnfv0   pharmacy_app_frontend.3      192.168.1.10:5000/pharmacy_frontend:2.0   worker1   Running         Running about a minut
e ago
i3wdjs60yr14   pharmacy_app_mongodb.1       mongo:8                                     manager   Running         Running about a minut
e ago
dpp5y90wq4ob   pharmacy_app_spring-boot.1   192.168.1.10:5000/pharmacy_backend:5.0    worker1   Running         Running about a minut
e ago
9x1nblwla8p0   pharmacy_app_spring-boot.2   192.168.1.10:5000/pharmacy_backend:5.0    worker2   Running         Running about a minut
e ago
9s5mcmk6kdph   pharmacy_app_spring-boot.3   192.168.1.10:5000/pharmacy_backend:5.0    manager   Running         Running about a minut
e ago
```

- Truy cập thử:



Docker Swarm Deploy Errors an
Dược An Khang - Hệ Thống Qu
Ask Gemini
Not secure
https://192.168.1.10/login
HỆ THỐNG AN KHANG
Đăng nhập cổng thông tin nhân viên
MÃ NHÂN VIÊN
Nhập mã nhân viên...
MẬT KHẨU
Nhập mật khẩu...
Đăng Nhập Hệ Thống
Search
ENG
10:40 PM
7/10/2026